

// from binary to microprocessors

Digital Electronics

A complete visual guide to logic gates, circuits, memory, and the building blocks of modern computing.

7

Chapters

1950s

Origins

2T+

Transistors / chip

v2026.01

Number Systems

All digital systems speak in binary
– but engineers use hex and octal
as shorthand to manage complexity.

BINARY – BASE 2

Digits: 0, 1 · Foundation of all digital logic

0101

HEXADECIMAL – BASE 16

Digits: 0–9, A–F · Used in memory addressing

0xFF

OCTAL – BASE 8

Digits: 0–7 · Common in Unix file permissions

0o77

Logic Gates

The fundamental decision-making units of every digital circuit

AND

$$A \cdot B = Y$$

Output is HIGH only when ALL inputs are HIGH.

NAND

$$(A \cdot B)' = Y$$

Universal gate — any circuit can be built from NAND alone.

OR

$$A + B = Y$$

Output is HIGH when ANY input is HIGH.

NOR

$$(A + B)' = Y$$

Also universal — the other complete logic gate set.

NOT

$$A' = Y$$

Inverts the input. HIGH becomes LOW and vice versa.

XOR

$$A \oplus B = Y$$

HIGH only when inputs differ. Core of adder circuits.

Combinational Circuits

Output depends only on the current input
– no memory, no state. Pure logic in action.

Multiplexer (MUX)

Selects one of N inputs to route to output

Half & Full Adder

Arithmetic addition using XOR and AND gates

Decoder & Encoder

Convert between binary codes and data lines

Sequential Circuits

Unlike combinational circuits, sequential circuits have memory. Output depends on both current inputs and past state – driven by a clock signal.

FLIP-FLOP TYPES

SR Flip-Flop

Set-Reset latch – basic memory element

S, R

D Flip-Flop

Data latch – stores one bit on clock edge

D

JK Flip-Flop

Resolves SR conflict – toggles when $J=K=1$

J, K

T Flip-Flop

Toggle – used in frequency dividers and counters

T

Memory & Storage

Where bits persist – volatile, non-volatile, fast, and slow

Registers

~1 cycle · bytes

Inside the CPU —
fastest possible,
smallest capacity

Cache

1–10 ns · MB

L1/L2/L3 SRAM —
bridges CPU and main
memory

RAM

50–100 ns · GB

DRAM — volatile
working memory,
loses data on power
off

Storage

ms · TB

SSD/HDD NAND Flash
— non-volatile long-
term storage

Real-World Applications

Digital electronics underpins every piece of technology built in the last 70 years. From pocket calculators to neural processing units – all from the same fundamental gates.

10B+

Digital chips shipped globally per year

Microprocessors & CPUs

Billions of logic gates executing instructions per second

FPGAs & Embedded Systems

Reconfigurable hardware for automotive, robotics, IoT

Digital Signal Processing

Audio, video, radar, 5G communication systems

AI Accelerators

Tensor cores and NPUs built from custom digital logic